

Attention Is All You Need

Abstract

A novel neural network architecture based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. The Transformer model achieves superior results on machine translation tasks.

Key Concept: Self-Attention

- Query (Q), Key (K), Value (V) matrices computed from input
- $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) * V$
- Allows each word to attend to every other word in the sequence

Architecture

- Encoder: 6 identical layers, each with multi-head attention + feed-forward
- Decoder: 6 identical layers with masked self-attention
- Multi-Head Attention: 8 parallel attention heads
- Positional Encoding: sine/cosine functions to encode word order

Results

- BLEU score: 28.4 on WMT 2014 English-to-German
- BLEU score: 41.0 on WMT 2014 English-to-French
- Training time: 12 hours on 8 P100 GPUs

Key Takeaway

Attention mechanisms alone are sufficient for sequence modelling. No RNN or CNN needed. Enables full parallelisation during training.